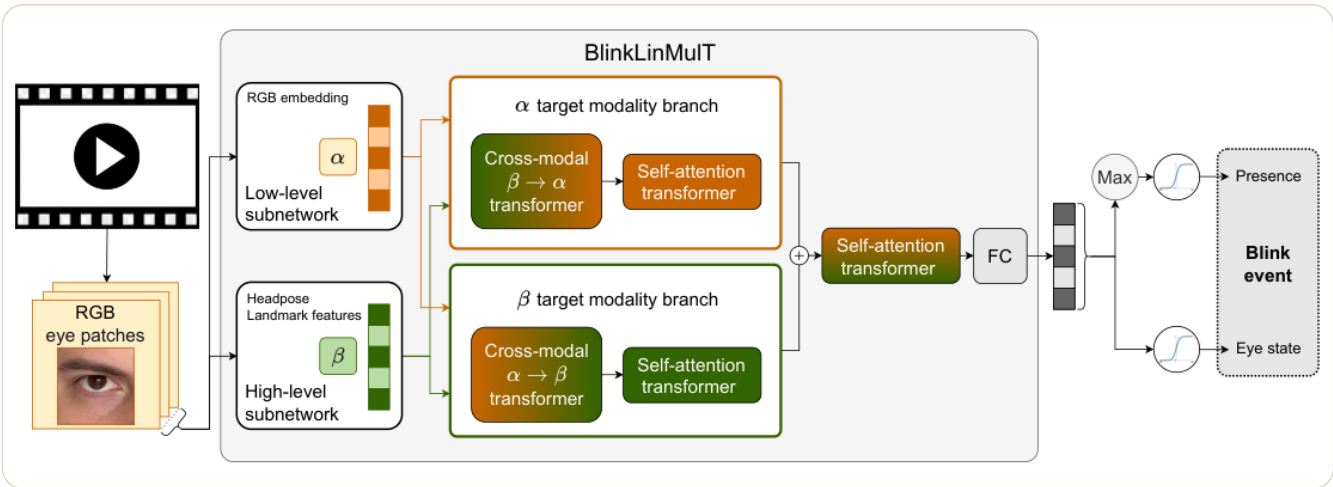


BlinkLinMulT: Transformer-Based Eye Blink Detection

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https://adamfodor.com/pdf/2023_Fodor_Adam_MDPI_BlinkLinMulT.pdf



BlinkLinMulT: a multimodal transformer architecture for blink presence detection and frame-wise eye state recognition.

Abstract

This work presents BlinkLinMulT, a transformer-based framework for eye blink detection. While most existing approaches rely on frame-wise eye state classification, recent advancements in transformer-based sequence models have not been explored in the blink detection literature. Our approach effectively combines low- and high-level feature sequences with linear complexity cross-modal attention mechanisms and addresses challenges such as lighting changes and a wide range of head poses. Our work is the first to leverage the transformer architecture for blink presence detection and eye state recognition while successfully implementing an efficient fusion of input features. In our experiments, we utilized several publicly available benchmark datasets (CEW, ZJU, MRL Eye, RT-BENE, EyeBlink8, Researcher's Night, and TalkingFace) to extensively show the state-of-the-art performance and generalization capability of our trained model. We hope the proposed method can serve as a new baseline for further research.

Task

Eye Blink Detection for Safety and Security

- Detecting eye blinks solves facial analysis tasks like deep fake detection and driver drowsiness monitoring.
- Methods are categorized into feature-based, appearance-based, and motion-based approaches.
- The approach is evaluated on blink presence detection and eye state recognition tasks using multiple public benchmark datasets.

Challenges

Challenges in Handling Noisy Data and Short Blinks

- Existing face recognition and frame-wise classification methods struggle with short blinks or noisy data.
- The CEW dataset lacks pre-defined splits, requiring a 10-fold cross-validation scheme to measure performance.
- Extreme head pose angles cause most errors due to eyelid visibility and landmark precision issues.

Proposed method

BlinkLinMulT: Linear Attention for Multi-Modal Input

- Modified multi-modal transformer considers RGB texture, iris and eye landmarks, ear, and head pose angles.
- Cross-modal transformers translate between landmark features and RGB texture embeddings.
- Linear multimodal transformer estimates blinks for each eye independently.

Main results

Superior F1 Scores on RT-BENEimg and TalkingFace

- ResNet50 achieved a 0.912 F1 score on RT-BENEimg, outperforming DenseNet121's 0.883.
- Training on the union of datasets improved F1 scores: CEW rose to 0.997, ZJU to 0.933, and RT-BENEimg to 0.913.
- A model achieved a perfect 1.0 F1 score on the TalkingFace dataset for blink presence detection.